COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	08 - 26 - 26	**DATE SUBMITTED:**	08 - 26 - 26
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PRELIM EXAM

1. PROCEDURE OUTPUT

Algorithm in Stack Array

Step 1: Create #include <iostream>, #include <stack> and define the maximum number to be implemented

Step 2: Initialize choice, maxcap and top to be used.

Step 3: When you integrate top, -1 will be the nullptr, it means it will be after the tail, where it can be a deciding factor whether if the stack is Full or Empty.

Step 4: Symmetric, isEmpty, and isFull will be bool. It is to show true or false when asked.

Step 5: push, pop, Top, and displayStack will be as define void.

Step 6: type int main, and brackets, this is will be used for later.

Step 7: Define bool Symmetric, return true if the top is the same as the tail, it should check if the whole array that is input by the user is symmetric. Return false if the array is not symmetric.

Step 8: Define bool isEmpty, use if loop, if the top is equal to -1, it means it is empty, if it is empty it will return true, if not return false.

Step 9: Define bool isFull, use if loop, if the top has reached the input number given, for example 10, the user cannot push or add more numbers.

Step 10: Define void push, it will use isFull, if the stack has reached the maxcap, it will print “stack overflow” and return, but if it is not full, it should let the user enter newdata.

Step 11: Define void pop, this will use isEmpty, If the stack is empty, it will print “the stack is underflow” if not, it should let the user pop /delete a number .

Step 12: Define void Top ,in checking the array, if the stack is empty, it will print “The stack is empty” if not it will print the latest number the user have push/inserted.

Step 13: Define void displayStack, if the stack is empty it will print “the stack is empty” if not it will display all the numbers that have been push/inserted, the pop/delete is not to be printed.

Step 14: going back to int main, here it will be designed as you want.

Step 15: Let the user pick from 1-7, 1. Push, 2. Pop , 3.Top, 4. Check (symmetric or not), 5. isEmpty, 6. isFull, 7. Display.

Step 16: case 1: push(); break; this will let the user insert numbers.

Step 17: case 2: pop(); break; this will let the user delete the input numbers.

Step 18: case 3: Top(); break; this will let the user see the latest number being pushed.

Step 19: case 4: Symmetric(); break; This will print true or false. this will let the user know if the array of number inputted is a stack of symmetric numbers, the code will check if the numbers are symmetric

Step 20: case 5: isEmpty(); break; This will print true or false, it will print false if the stack has still numbers on it, true is it is empty.

Step 21: case 6: isFull() break; This will print true or false, it will print false if the stack hasnt reached the maxcap given, and if yes true.

Step 22: case 7: displayStack; break; This will print all the input numbers of the user, and will print “the stack is empty” if the stack is empty.

Analysis:
This algorithm is a brief step by step on how to do the code, it will let you think and do how to do the array stack that I have in mind. It has the basics and the thought to complete the task. It just needed to be researched or have the idea to do the activity. The idea I want to implement is a simple code, where the user will choose from the choices and be able to push, pop, check and know whether, what is the latest number that has been input, If the stack is empty or full, and if the numbers given is symmetric. The symmetric part, it should check the whole array, if the first one and the last one and the ones that will continue them are the same element as them. If not it should return as false.

```
int main(){
    while (true){
        std::cout << "========================================\n";
        std::cout << "Stack Operations\n";
        std::cout << "1. PUSH 2. POP 3. TOP 4. Check (Symmetric or Not) 5. isEmpty 6. isFull 7. Display\n";
        std::cout << "========================================\n";
        std::cin >> choice;

        switch(choice){
            case 1 : push();
            break;
            case 2 : pop();
            break;
            case 3 : Top();
            break;
            case 4 : std::cout << "is the line up symmetric? " << Symmetric() << std::endl;
            break;
            case 5 : std::cout << "is stack empty? " << isEmpty() << std::endl;
            break;
            case 6: std::cout << "is the stack full? " << isFull () <<std::endl;
            break;
            case 7: displayStack() ;
            break;
            default: std::cout<<"Invalid Choice" << std::endl;
            break;
        }
```

```
bool Symmetric() {
    if (top = 5) return true;
    while (top != 5) return false;
}
```

The code worked at first, I couldn't figured out why it couldn't be false if the elements were different. And I changed the code after a while, trying some things, that's why it should have an error now.

Algorithm in QUEUE

Step 1: Do a header file where you will code how you will print it in the .cpp file

Step 2: Do a class, the private will have the array, capacity, size, front and back

Step 3: the public will have bool, void of full, empty, size, clear, front, back,enqueue, dequeue, symmetric and array.

Step 4: Then define all that.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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